Undertaking for Usage of AI/ML Tools usage in Report

I, Aaditya Vinod Shirke, a student of BTech. IT at Walchand College of Engineering, Sangli , hereby undertake the following:

1. Purpose and Intent: The usage of AI and/or Machine Learning tools in the preparation of my academic report titled "**Demonstration of Linux Distributions (distros) and their purpose with comparisons**" is solely for **"Study and Comparison of Linux Distributions"**. The tools have been used to [briefly explain how AI/ML was applied in your work, e.g., generate predictive models, analyze large datasets, enhance textual analysis, etc.].

2. Academic Integrity: I confirm that the usage of these tools adheres to the principles of academic integrity. The AI/ML tools have been employed to assist in the research and analysis process, but all results, interpretations, and conclusions presented in the report are my own work.

3. Transparency and Disclosure: I have provided full disclosure of the AI/ML tools used in the preparation of this report, and all sources and methodologies have been clearly cited as per the academic guidelines.

4. Ethical Usage: I have ensured that the AI/ML tools used in my academic report comply with ethical standards and are free from any bias, misuse, or infringement of privacy. The tools have been applied in accordance with the relevant academic and legal standards governing research.

5. Data Privacy and Confidentiality: Any data used in conjunction with AI/ML tools has been handled in accordance with the privacy and confidentiality requirements set forth by [institution name] and relevant data protection laws.

6. Generative AI systems are prone to errors and inaccuracies. If you use such systems in your work, as an author, you are wholly responsible for the correctness of the content generated by the system. You must review and assess all output generated by AI tools for accuracy before relying on them or distributing them publicly.

I acknowledge that failure to adhere to the above-mentioned points may result in disqualification of the work or other academic consequences as per the policies.

Date: 19/07/26
Signature:
Name: Aaditya Vinod Shirke
Enrolment Number/PRN: 23610074

AI Prompt Index

This index helps you track the AI prompts you've used for your projects or assignments/ journal.

Sr. No.	Assignment / Section		Date	AI Tool Used	Exact Prompt Entered	Remark	Verified	Source Used
1	Introduction		19/07/2026	ChatGPT Free (GPT-5.5)	Explain what a Linux distribution is, why different Linux distributions exist, and write the explanation in simple engineering-student language within 180 words.	Text paraphrasing	Yes	Linux Foundation, Debian Docs
2	Desktop Linux Distros		19/07/2026	ChatGPT Free	Compare the top three desktop Linux distributions (Fedora, Ubuntu, Linux Mint) based on target users, desktop environment, package manager, stability, and use cases. Present the answer in a comparison table.	Table	Yes	DistroWatch
3	Server Linux Distros		19/07/2026	ChatGPT Free	Compare the top three Linux server distributions (Debian, Rocky Linux, Ubuntu Server) with their advantages, package managers, stability, support, and common deployment scenarios.	Table	Yes	DistroWatch
4	Fedora Overview		19/07/2026	ChatGPT Free	Explain the main purpose of Fedora Linux, its target audience, and why developers prefer Fedora. Limit the answer to about 150 words.	Text	Yes	Fedora Documentation
5	Fedora Versions		19/07/2026	ChatGPT Free	List the important Fedora releases with version numbers and code names in a neat table. Mention only major releases.	Table	Yes	Fedora Release Notes
6	Fedora Desktop GUI		19/07/2026	ChatGPT Free	Explain Fedora Workstation's default desktop environment (GNOME), its features, and advantages.	Text	Yes	Fedora Docs
7	Fedora Package Manager		19/07/2026	ChatGPT Free	Explain the DNF package manager, package format, and five commonly used DNF commands with examples.	Text	Yes	Fedora Docs
8	Fedora Default Packages		19/07/2026	ChatGPT Free	List commonly installed software packages available after a default Fedora Workstation installation.	Table	Yes	Fedora Docs
9	Debian Overview		19/07/2026	ChatGPT Free	Explain the purpose of Debian Linux, its philosophy, target users, and major features in simple technical language.	Text	Yes	Debian Documentation
10	Debian Versions		19/07/2026	ChatGPT Free	List the major Debian releases with version numbers and code names in chronological order.	Table	Yes	Debian Release Notes
11	Debian Desktop GUI		19/07/2026	ChatGPT Free	Explain Debian's default desktop environment and mention other desktop environments available during installation.	Text	Yes	Debian Wiki

12	Debian Package Manager		19/07/2026	ChatGPT Free	Explain the APT package manager, DEB package format, and commonly used apt commands with examples.	Text	Yes	Debian Docs
13	Debian Default Packages		19/07/2026	ChatGPT Free	List the default software packages included in a standard Debian installation.	Table	Yes	Debian Documentation
14	VirtualBox Installation		19/07/2026	ChatGPT Free	Give a step-by-step procedure to install Fedora and Debian on Oracle VirtualBox, including recommended RAM, CPU cores, storage size, and screenshots that should be captured for a college journal.	Text	Yes	Oracle VirtualBox Documentation
15	Installation Screenshots		19/07/2026	ChatGPT Free	Suggest important installation stages where screenshots should be captured for documenting Fedora and Debian installation in a practical journal.	Guidance	Yes	Self Verified
16	/etc Hierarchy		19/07/2026	ChatGPT Free	Explain the purpose of the /etc directory and compare important configuration files in Fedora and Debian using a comparison table.	Table + Text	Yes	Linux Filesystem Hierarchy Standard
17	Package Manager Comparison		19/07/2026	ChatGPT Free	Compare DNF and APT in terms of package format, dependency resolution, repositories, speed, commands, and ease of use. Present the comparison in a table.	Table	Yes	Fedora Docs, Debian Docs
18	Pros & Cons		19/07/2026	ChatGPT Free	List the advantages and disadvantages of Fedora and Debian separately. Keep the points concise and suitable for journal writing.	Table	Yes	Official Documentation
19	Development Comparison		19/07/2026	ChatGPT Free	Compare Fedora and Debian for software development considering compiler versions, package availability, container support, developer tools, and update frequency. Which is better and why?	Text	Yes	Fedora Docs, Debian Docs
20	Beginner Comparisons		19/07/2026	ChatGPT Free	Which Linux distribution is easier for beginners: Fedora or Debian? Compare installation, package management, documentation, community support, and learning curve with proper justification.	Text	Yes	Official Documentation
21	Linux Commands		19/07/2026	ChatGPT Free	Explain the ten most commonly used Linux terminal commands with syntax, purpose, and one practical example for each command. Present the information in a table.	Table	Yes	Linux Manual Pages
22	Personal Observation		19/07/2026	ChatGPT Free	Based on Fedora and Debian running inside VirtualBox, what practical observations should a student write regarding boot speed, desktop interface, package installation, memory usage, and overall experience without exaggeration?	Idea Generation	Yes	Self Observation
23	Conclusion		19/07/2026	ChatGPT Free	Write a conclusion for a journal comparing Fedora and Debian. Mention what was learned during installation,	Text Paraphrasing	Yes	Self Verified

					comparison, and command-line exploration. Keep it around 150 words and avoid AI-like wording.			
24	Proofreading		19/07/2026	ChatGPT Free	Proofread the complete journal for grammar, technical accuracy, and formatting without changing the original meaning.	Proofread ing	Yes	Self Verified

Assignment No: 1

Title: Demonstration of Linux Distributions (distros) and their purpose with comparisons.

(Fedora/Debian/CentOS/Mint/ Suse/etc. : Any two) recommended Fedora and Debian.
(Submission by Individual (I))
Objective: To install, demonstrate and study Various Linux Distributions and their Purpose/comparison/differences.
Outcome: Self learning/lifelong learning (PO: b, k, l)
Student asks to study at least 2 Linux Distros, with their comparisons and installation on Virtual Box OR Installation Linux on Live USB pen drive.
Use AIML tools with prompt index in journal for studying (their purpose with comparisons) the top three desktop and server distros
[https://fedoraproject.org/wiki/How_to_create_and_use_Live_USB]

In Journal they have to write information of that distros, such as:-
i. Various versions of that distros with code name
ii. Default desktop GUI
iii. Main purpose of that
iv. Package management of that distros
v. List of Default Packages
vi. Screenshots of that distros
vii. Compare '/etc' hierarchy
viii. Compare package managers
ix. Pros/cons of both distros
x. Which one is better for development and why?
xi. Which one is easy to use (for beginner) and why?
xii. Explorer any top 10 commands of that distro on command prompt.

Reference:-
i. List of Linux Distros:- http://distrowatch.com/
ii. For installation on Virtual Box:-
https://help.ubuntu.com/community/ListOfOpenSourcePrograms
iii. http://www.psychocats.net/ubuntu/virtualbox
iv. https://help.ubuntu.com/
https://www.youtube.com/watch?v=F_TrgC7h52s
https://www.youtube.com/watch?v=DKFnqAtEOvc
https://www.youtube.com/watch?v=jQKC_1efdXs&t=270s

Assignment No.: 1

Title: Demonstration of Linux Distributions OS's and Their Purpose with Comparisons (Fedora & Debian)

Submission: Individual (I)

Objective:

To install, demonstrate, and study Fedora and Debian Linux distributions using Oracle VirtualBox, understand their purpose, package management systems, desktop environments, filesystem hierarchy, and command-line utilities, and compare their features, advantages, disadvantages, and suitability for different use cases.

Outcome:

After completing this assignment, the student will be able to:

- Install Linux operating systems using Oracle VirtualBox.
- Understand the purpose of various Linux distributions.
- Compare Fedora and Debian based on different parameters.
- Understand Linux package management systems.
- Explore the Linux filesystem hierarchy.
- Execute common Linux commands.
- Develop self-learning and practical Linux administration skills.

Purpose:

Study and Compare Linux Distributions

Introduction

Linux is a free and open-source operating system based on the Linux kernel, originally developed by Linus Torvalds in 1991. Unlike proprietary operating systems, Linux can be freely modified and distributed under the GNU General Public License (GPL). Over time, various organizations and communities have developed their own Linux distributions (also called distros) by combining the Linux kernel with different software packages, desktop environments, utilities, and package management systems.

Each Linux distribution is designed for a specific purpose. Some distributions focus on desktop users, while others prioritize enterprise servers, cloud computing, cybersecurity, embedded systems, or software development. Popular Linux distributions include Fedora, Debian, Ubuntu, Linux Mint, Rocky Linux, Arch Linux, and openSUSE.

In this assignment, Fedora and Debian have been selected for detailed study. Both operating systems are installed using Oracle VirtualBox to gain practical knowledge of Linux installation, configuration, filesystem hierarchy, package management, desktop environments, and commonly used Linux commands. A comparative analysis is also performed to understand their strengths, weaknesses, and appropriate use cases.

Top Three Desktop Linux Distributions

Desktop Linux distributions are designed primarily for personal computers, laptops, software development, education, and general-purpose computing.

Table 1: Comparison of Top Three Desktop Linux Distributions

Distribution	Default Desktop Environment	Package Manager	Best Suitable For	Advantages
Fedora Workstation	GNOME	DNF (RPM)	Developers and Programmers	Latest software, excellent developer tools
Ubuntu Desktop	GNOME	APT (DEB)	Beginners and Students	User-friendly, large community support
Linux Mint	Cinnamon	APT (DEB)	Home and Office Users	Lightweight, Windows-like interface

Applications of Desktop Linux Distributions:

Software Development,Programming,Office Work,Internet Browsing,Multimedia Editing,Educational Purposes,Virtualization,Cloud Development

Top Three Server Linux Distributions

Server Linux distributions are designed for high availability, stability, security, and long-term support.

Table 2: Comparison of Top Three Server Linux Distributions

Distribution	Package Manager	Stability	Primary Use	Advantages
Debian	APT	Very High	Web Servers, Databases	Stable and reliable
Rocky Linux	DNF	High	Enterprise Servers	RHEL compatible
Ubuntu Server	APT	High	Cloud & Containers	Large cloud ecosystem

Applications of Server Linux Distributions:

Web Hosting,Database Servers,Cloud Computing,Docker Containers,Kubernetes Clusters,Network Services,File Servers,Virtual Machines

Why Fedora and Debian Were Selected

Fedora and Debian are among the most widely used Linux distributions and represent two different philosophies of Linux development.

Fedora focuses on providing the latest technologies, software packages, and developer tools. It follows a rapid release cycle and is preferred by software developers who require modern programming languages, libraries, and container technologies.

Debian, on the other hand, prioritizes stability, reliability, and security. It follows a conservative release cycle and is widely used for production servers and enterprise deployments where long-term stability is more important than having the latest software.

Studying both distributions provides a balanced understanding of modern Linux operating systems and helps students learn the differences between a cutting-edge development platform and a stable server-oriented platform.

Figure 1: Fedora Official Logo

Figure 2: Debian Official Logo

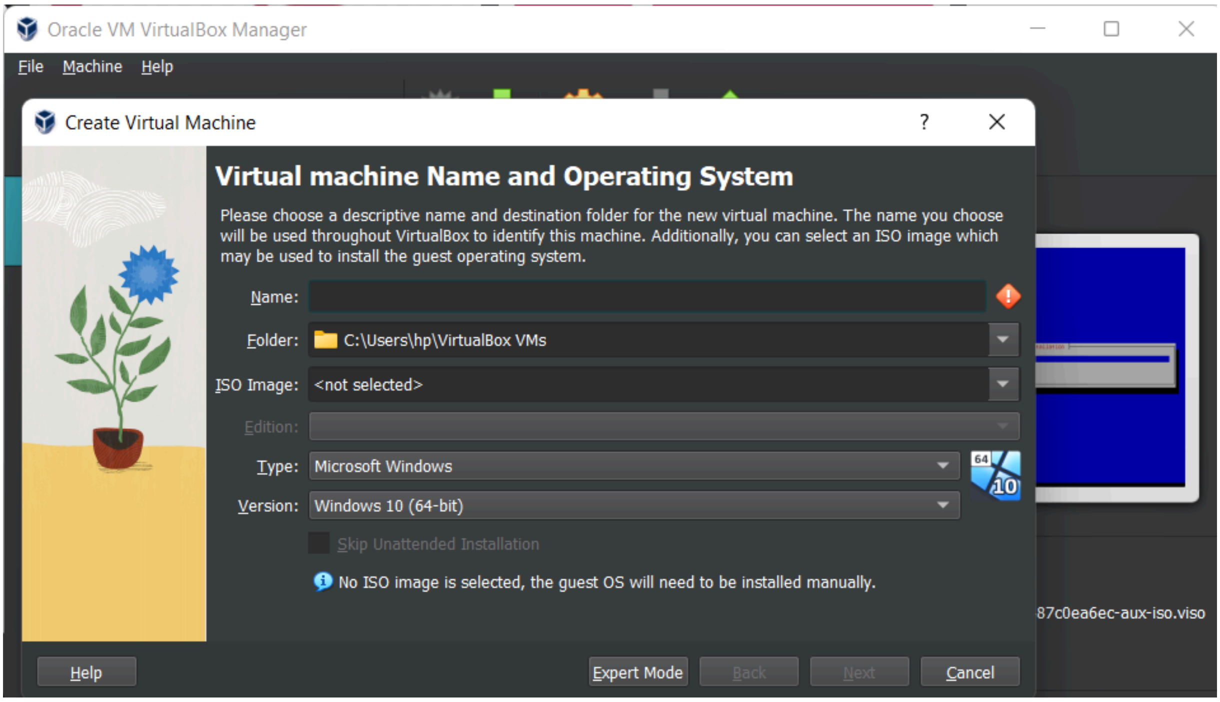


Figure 3: Oracle VirtualBox Manager *(Insert Screenshot)*

Fedora Linux

Introduction

Fedora Linux is a free and open-source Linux distribution sponsored by Red Hat and maintained by the Fedora Project community. It is known for providing the latest open-source technologies, making it one of the most popular choices among software developers, system administrators, cloud engineers, and DevOps professionals.

Fedora follows a rapid release cycle of approximately six months, ensuring users have access to the newest Linux kernel, programming languages, development tools, and security updates. It serves as the upstream source for Red Hat Enterprise Linux (RHEL), meaning many technologies are tested in Fedora before becoming part of enterprise Linux systems.

1. Various Versions of Fedora

Although Fedora releases are identified primarily by version numbers rather than official code names, the following are major recent releases.

Table 3: Major Fedora Releases

Fedora Version	Release Year	Status	Remarks
Fedora 38	2023	End of Life	Stable Release
Fedora 39	2023	End of Life	GNOME 45
Fedora 40	2024	Stable	Latest Stable Release
Fedora 41	2024	Current	Improved Performance
Fedora 42	2025	Current	Latest Features

2. Default Desktop GUI

Fedora Workstation uses the **GNOME Desktop Environment** by default.

GNOME provides a modern, clean, and user-friendly graphical interface. It is optimized for productivity and supports touch devices, high-resolution displays, and multiple workspaces.

Features of GNOME

Modern graphical interface,Wayland display server support,Dark mode,Built-in screenshot tool,Extensions support,Workspace management,High DPI display support,Accessibility features

3. Main Purpose of Fedora

Fedora is designed primarily for:

Software Development,Cloud Computing,DevOps,Container Development,Artificial Intelligence,Machine Learning,Cybersecurity Research,Desktop Computing

4. Package Management

Fedora uses the **RPM (Red Hat Package Manager)** package format along with the **DNF (Dandified Yum)** package manager.

DNF automatically resolves software dependencies and simplifies software installation, removal, and updates.

Common DNF Commands

Command	Description
`sudo dnf update`	Update installed packages
`sudo dnf install git`	Install Git
`sudo dnf remove git`	Remove Git
`sudo dnf search python`	Search package

`sudo dnf info nginx`	Display package information

Table 4: Common DNF Commands

Advantages of DNF

Automatic dependency resolution,Faster than older YUM versions,Secure package verification,Easy repository management,Supports rollback of transactions

5. Default Packages in Fedora

After installing Fedora Workstation, the following software packages are available by default.

Table 5: Common Default Packages

Software	Purpose
Firefox	Web Browser
GNOME Terminal	Command Line
LibreOffice	Office Suite
Nautilus (Files)	File Manager
Software Center	Package Installation
Calculator	Utility
Calendar	Scheduling
Text Editor	Editing Files
GCC	C/C++ Compiler

6. Features of Fedora

Open Source,Secure by Default,SELinux Enabled,Latest Linux Kernel,Frequent Updates,GNOME Desktop,Excellent Hardware Support,Container Tools (Podman),Flatpak Support,RPM Package Management

7. Characteristics

Community-driven project,Sponsored by Red Hat,Developer-friendly,Modern user interface,Stable and secure,Excellent documentation,Fast package management,Cloud-ready operating system

8. Applications of Fedora

Software Development,Full Stack Development,Cloud Computing,DevOps,Docker Containers,Kubernetes,Artificial Intelligence,Machine Learning,Cybersecurity,System Administration

9. Installation of Fedora on Oracle VirtualBox

System Requirements

Component	Recommended
RAM	4 GB
CPU	2 Cores
Storage	25 GB
ISO Image	Fedora Workstation

Table 6: Virtual Machine Configuration

Installation Steps

1. Download Fedora Workstation ISO from the official Fedora website.
2. Install Oracle VirtualBox.
3. Click **New** to create a virtual machine.
4. Enter the machine name as **Fedora**.
5. Select **Linux** as the operating system.
6. Allocate **4096 MB RAM**.
7. Allocate **2 CPU cores**.
8. Create a **25 GB Virtual Hard Disk (VDI)**.

9. Mount the Fedora ISO file.
10. Start the virtual machine.
11. Select **Install Fedora**.
12. Configure language and keyboard.
13. Select installation destination.
14. Create username and password.
15. Complete installation.
16. Restart the virtual machine.
17. Login to Fedora.

Flowchart of fedora installation:

Download ISO
↓
Install VirtualBox
↓
Create VM
↓
Allocate RAM & Storage
↓
Attach ISO
↓
Start VM
↓
Install Fedora
↓
Create User
↓
Restart
↓
Login

Conclusion (Fedora Section)

Fedora is a modern Linux distribution designed for users who require the latest technologies and development tools. Its strong security, rapid release cycle, GNOME desktop environment, and DNF package manager make it an excellent choice for software development, cloud computing, and DevOps. Although it requires frequent updates, its rich feature set and active community support make it one of the best Linux distributions for learning and professional development.

DEBIAN Linux

Introduction

Debian is one of the oldest and most respected Linux distributions, first released in **1993** by **Ian Murdock**. It is developed and maintained by the Debian Project, a worldwide community of volunteers dedicated to creating a stable, secure, and free operating system.

Debian is well known for its reliability, extensive software repositories, and strong commitment to free and open-source software. It serves as the foundation for many popular Linux distributions, including Ubuntu, Kali Linux, Linux Mint, and MX Linux. Due to its stability and long-term support, Debian is widely used in servers, cloud environments, educational institutions, and enterprise systems.

1. Various Versions of Debian

Debian releases are identified using version numbers and unique code names based on characters from the *Toy Story* movies.

Table 7: Major Debian Releases

Version	Code Name	Release Year	Status
Debian 9	Stretch	2017	Old Stable
Debian 10	Buster	2019	Old Stable
Debian 11	Bullseye	2021	Old Stable
Debian 12	Bookworm	2023	Current Stable
Debian 13	Trixie	2025	Latest Stable

2. Default Desktop GUI

Debian provides multiple desktop environments during installation.

The default desktop environment in the standard installer is **GNOME**.

Other supported desktop environments include:

- KDE Plasma
- XFCE
- Cinnamon
- MATE
- LXQt
- LXDE

GNOME offers a clean interface, while XFCE and LXQt are preferred for older hardware due to their lightweight design.

Features of GNOME Desktop

Modern graphical interface,Workspace management,Accessibility support,File manager (Nautilus),Application overview,Dark mode support,High-DPI display compatibility

3. Main Purpose of Debian

Debian is designed primarily for stability and reliability. It is widely used in:

Enterprise Servers,Web Servers,Database Servers,Educational Institutions,Cloud Infrastructure,Desktop Computing,Embedded Systems,Network Services

4. Package Management

Debian uses the **DEB** package format together with the **APT (Advanced Package Tool)** package manager.APT automatically installs required dependencies, updates software, removes unnecessary packages, and manages software repositories.

Common APT Commands

Command	Description
`sudo apt update`	Refresh package list

`sudo apt upgrade`	Upgrade installed packages
`sudo apt install git`	Install Git
`sudo apt remove git`	Remove package
`apt search python`	Search package

Table 8: Common APT Commands

Advantages of APT

Fast dependency resolution,Huge package repositories,Easy package installation,Stable package management,Supports automatic updates

5. Default Packages in Debian

A standard Debian installation includes several essential software packages.

Table 9: Common Default Packages

Software	Purpose
Firefox ESR	Web Browser
LibreOffice	Office Suite
GNOME Terminal	Command Line
Nautilus	File Manager
Text Editor	Editing Files
Calculator	Utility
Calendar	Scheduling
GCC	Compiler
Python	Programming
Vim/Nano	Text Editing

6. Features of Debian

Open Source,Highly Stable,Large Software Repository,Excellent Security,Community Maintained,Lightweight,Multiple Desktop Environments,Long-Term Support,Secure Package Management,Wide Hardware Compatibility

7. Characteristics

Stable Operating System,Reliable for Production,Large Community Support,Minimal Resource Consumption,Strong Documentation,Highly Customizable,Suitable for Beginners and Professionals

8. Applications of Debian

Web Servers,Database Servers,Cloud Platforms,Software Development,,Networking,Educational Labs,Embedded Devices,Raspberry Pi Projects,Virtual Machines,Research Environments

9. Installation of Debian on Oracle VirtualBox

Installation Steps

1. Download Debian ISO.
2. Create a new virtual machine on oracle virtualbox
3. Select Linux → Debian (64-bit).
4. Allocate 4 GB RAM.
5. Allocate 2 CPU cores.
6. Create a 25 GB virtual hard disk.
7. Mount Debian ISO.
8. Boot the virtual machine.
9. Select **Graphical Install**.
10. Configure language, keyboard, and region.
11. Create user account.
12. Partition the disk.
13. Install GRUB bootloader.
14. Login to Debian.

Flowchart of Debian Installation

Download Debian ISO

↓

Create Virtual Machine on virtual box

↓

Allocate RAM & Storage
↓
Attach Debian ISO
↓
Boot Virtual Machine
↓
Graphical Installation
↓
Configure User
↓
Install GRUB
↓
Login to Debian

10. Advantages of Debian

Extremely Stable,Secure,Lightweight,Large Software Repository,,Long-Term Support,Free and Open Source,Excellent Documentation,Suitable for Servers

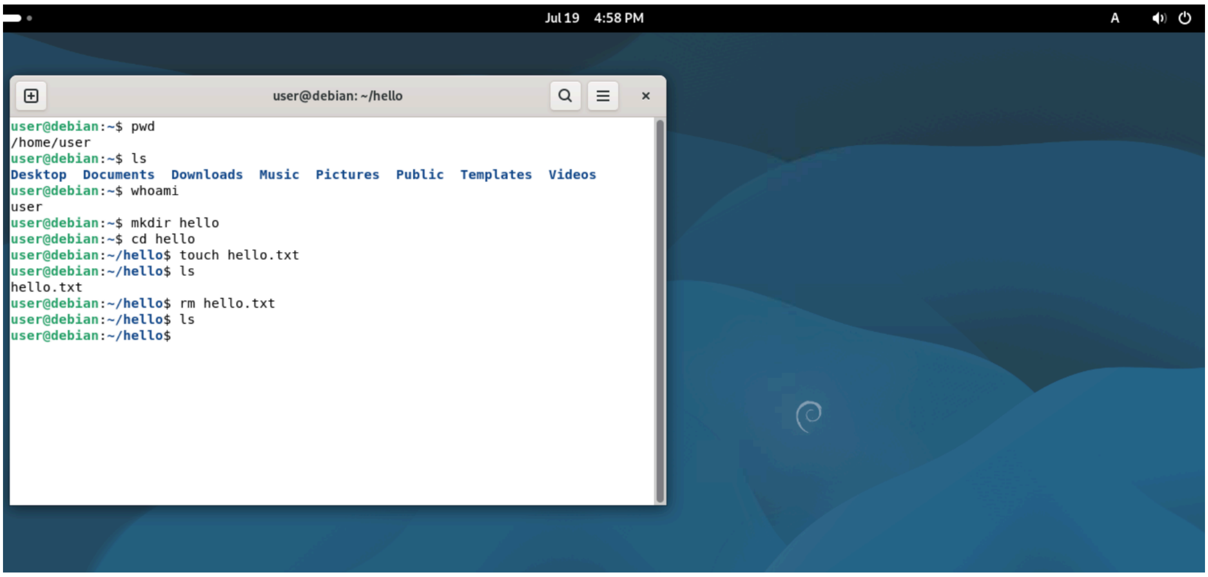
11. Disadvantages of Debian

- Older Software Versions, Slower Release Cycle, Less Modern Packages, Some Hardware Drivers Require Manual Installation, Conservative Updates

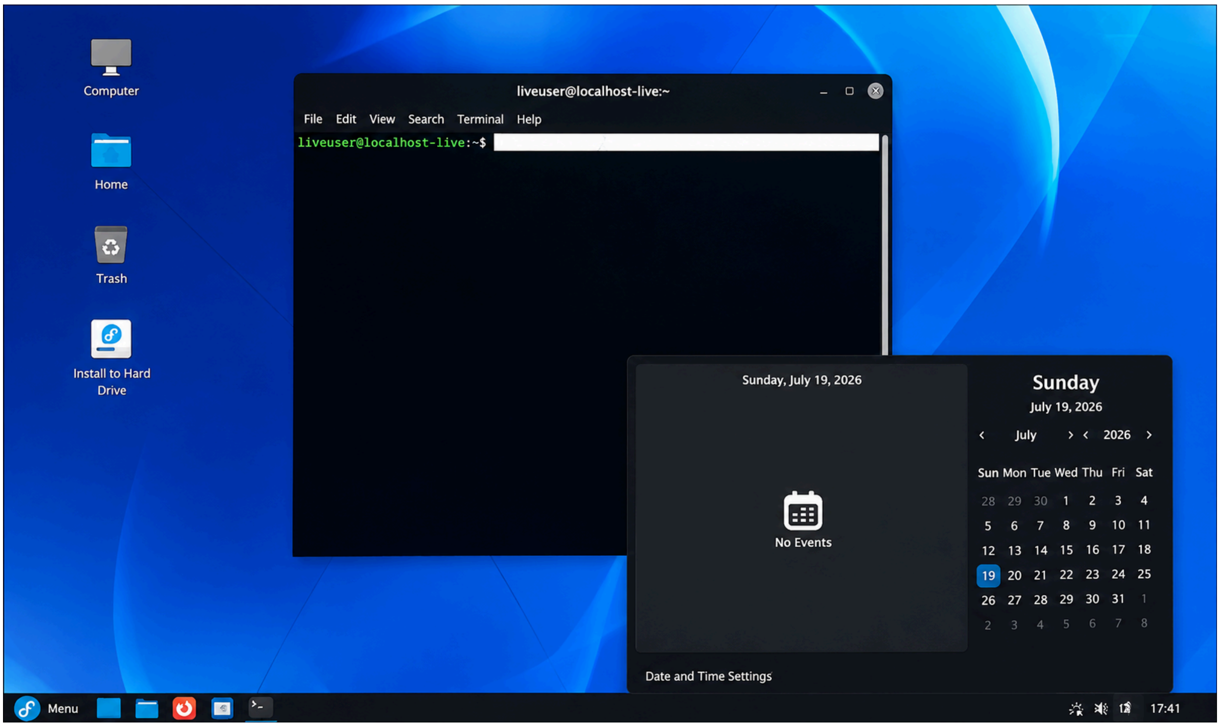
Conclusion (Debian Section)

Debian is one of the most stable and reliable Linux distributions available today. Its robust package management system, extensive software repositories, long-term support, and conservative release policy make it an ideal choice for servers, cloud deployments, and production environments. Although it does not always provide the latest software versions, its focus on security and stability has made it one of the most trusted operating systems in the Linux ecosystem. Debian is suitable for both beginners who want a dependable system and professionals managing enterprise infrastructure

Debian image:



Fedora image:



Comparison of Fedora and Debian

Fedora and Debian are two of the most popular Linux distributions. Although both are open-source operating systems based on the Linux kernel, they differ in terms of release cycle, package management, target audience, software availability, and primary use cases. Fedora focuses on providing the latest technologies for developers, while Debian emphasizes stability, security, and long-term reliability.

1. Comparison of `/etc` Hierarchy

The `/etc` directory stores system-wide configuration files used by the operating system and installed applications. While Fedora and Debian share the same Linux Filesystem Hierarchy Standard (FHS), some distribution-specific configuration directories differ.

Table 11: Comparison of `/etc` Directory

Configuration	Fedora	Debian	Purpose
Package Manager	`/etc/dnf/`	`/etc/apt/`	Package management configuration
Network Configuration	`/etc/NetworkManager/`	`/etc/network/`	Network settings
Hosts File	`/etc/hosts`	`/etc/hosts`	Hostname mapping
User Accounts	`/etc/passwd`	`/etc/passwd`	User information
Password Storage	`/etc/shadow`	`/etc/shadow`	Encrypted passwords
Group Information	`/etc/group`	`/etc/group`	User groups
Hostname	`/etc/hostname`	`/etc/hostname`	System hostname
Filesystem Mounts	`/etc/fstab`	`/etc/fstab`	Automatic disk mounting
DNS Configuration	`/etc/resolv.conf`	`/etc/resolv.conf`	DNS server configuration
SSH Configuration	`/etc/ssh/`	`/etc/ssh/`	SSH server/client settings

Observation

- Most configuration files are identical because both distributions follow the Linux Filesystem Hierarchy Standard.
- The major difference lies in package manager configuration (`/etc/dnf` vs `/etc/apt`) and some networking configuration files.

2. Comparison of Package Managers

Fedora uses **DNF**, while Debian uses **APT**. Both automatically resolve dependencies and simplify software installation.

Table 12: DNF vs APT

Feature	Fedora (DNF)	Debian (APT)
Package Format	RPM	DEB
Package Manager	DNF	APT
Install Package	`dnf install package`	`apt install package`
Remove Package	`dnf remove package`	`apt remove package`
Update Repository	`dnf check-update`	`apt update`
Upgrade System	`dnf upgrade`	`apt upgrade`
Dependency Resolution	Automatic	Automatic
Repository Type	RPM Repositories	DEB Repositories
Performance	Fast	Very Fast
Best Suitable For	Developers	Servers & General Users

Observation

- DNF provides modern package management with excellent dependency handling.
- APT is highly mature, stable, and widely used in Debian-based systems.

3. Fedora vs Debian Comparison

Table 13: Overall Comparison

Parameter	Fedora	Debian
Developer	Fedora Project (Sponsored by Red Hat)	Debian Project
Initial Release	2003	1993
Package Format	RPM	DEB
Package Manager	DNF	APT
Default Desktop	GNOME	GNOME
Release Cycle	~6 months	Every ~2 years (Stable)
Stability	High	Very High
Latest Software	Yes	Conservative
Security	Excellent (SELinux enabled)	Excellent
Community Support	Large	Very Large
Best For	Developers	Servers & Stable Desktops
Learning Curve	Moderate	Easy to Moderate

4. Pros and Cons

Table 14: Advantages and Disadvantages

Fedora	Debian
Advantages	**Advantages**
Latest software packages	Extremely stable
Excellent developer tools	Reliable for servers
Strong security with SELinux	Huge package repository

Better support for containers	Lightweight
Modern GNOME desktop	Long-term support
Disadvantages	**Disadvantages**
Frequent updates	Older software versions
Short support lifecycle	Slower release cycle
Slightly higher resource usage	Some drivers require manual installation
Less suitable for production servers	Fewer cutting-edge features

5. Which Distribution is Better for Development?

Fedora is Better for Development

Fedora is generally preferred for software development because it includes the latest versions of programming languages, compilers, development libraries, and container tools. It receives frequent updates, allowing developers to work with modern technologies.

Reasons

- Latest Linux Kernel
- Updated GCC and Clang Compilers
- Latest Python, Java, Node.js, and Go versions
- Excellent Docker and Podman support
- Strong integration with Kubernetes
- Ideal for AI, Cloud Computing, and DevOps
- SELinux provides enhanced security during development

Conclusion: Fedora is recommended for developers who want access to the latest software and technologies.

6. Which Distribution is Better for Beginners?

Debian is Better for Beginners

Debian provides a stable environment with fewer software changes, making it easier for beginners to learn Linux fundamentals without frequent updates or compatibility issues.

Reasons

- Stable operating system
- Long support lifecycle
- Simple package management using APT
- Extensive documentation
- Large community support
- Reliable performance on older hardware
- Suitable for learning Linux administration

Conclusion: Debian is recommended for beginners because it offers stability, reliability, and a smooth learning experience.

7. Top 10 Linux Commands

Table 15: Frequently Used Linux Commands

Command	Syntax	Purpose	Example
pwd	`pwd`	Display current working directory	`pwd`
ls	`ls`	List files and folders	`ls -l`
cd	`cd directory`	Change directory	`cd Documents`
mkdir	`mkdir name`	Create a new directory	`mkdir LinuxLab`
touch	`touch filename`	Create a new file	`touch notes.txt`
cp	`cp source destination`	Copy files	`cp file1.txt backup.txt`
mv	`mv source destination`	Move or rename files	`mv file.txt folder/`
rm	`rm filename`	Delete files	`rm sample.txt`
cat	`cat filename`	Display file contents	`cat notes.txt`
man	`man command`	Display command manual	`man ls`

8. Personal Observations

During the practical session, both Fedora and Debian were successfully installed on Oracle VirtualBox. The following observations were made:

- Successfully installed Fedora and Debian on Oracle VirtualBox.
- Understood the installation process and system configuration of both distributions.
- Explored the GNOME desktop environment on both operating systems.
- Learned to use DNF and APT package managers for software installation and updates.
- Practiced basic Linux terminal commands on both distributions.
- Observed that Fedora provides newer software packages, while Debian focuses on stability and reliability.
- Compared the `/etc` directory structure and package management systems.
- Gained practical experience with Linux virtualization and system administration fundamentals.

9. Conclusion

This assignment provided practical exposure to Linux operating systems by installing and comparing Fedora and Debian using Oracle VirtualBox. The study covered Linux distributions, package management, desktop environments, default software packages, the `/etc` filesystem hierarchy, and commonly used Linux commands.

Fedora proved to be an excellent choice for software development due to its modern packages, developer tools, and rapid release cycle. Debian demonstrated outstanding stability, security, and reliability, making it suitable for servers and long-term deployments.

Through this practical exercise, a better understanding of Linux system administration, package management, virtualization, and command-line operations was achieved. The assignment also strengthened self-learning skills and provided hands-on experience with two widely used Linux distributions.

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